Thermodynamic description of the Cu–Pb system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Pb binary system as calculated with pycalphad from the SGTE dataset CuPb-86Hay-LB.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 → (Cu) + L2, lies at 956.8 °C.

1. Introduction

The Cu–Pb system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of F.H. Hayes, H.L. Lukas, G. Effenberg, G. Petzow, Z. Metallkd., 77, 749-54(1986). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Cu reproduced by the dataset is 1084.6 °C.

2. Phase diagram and invariant reactions

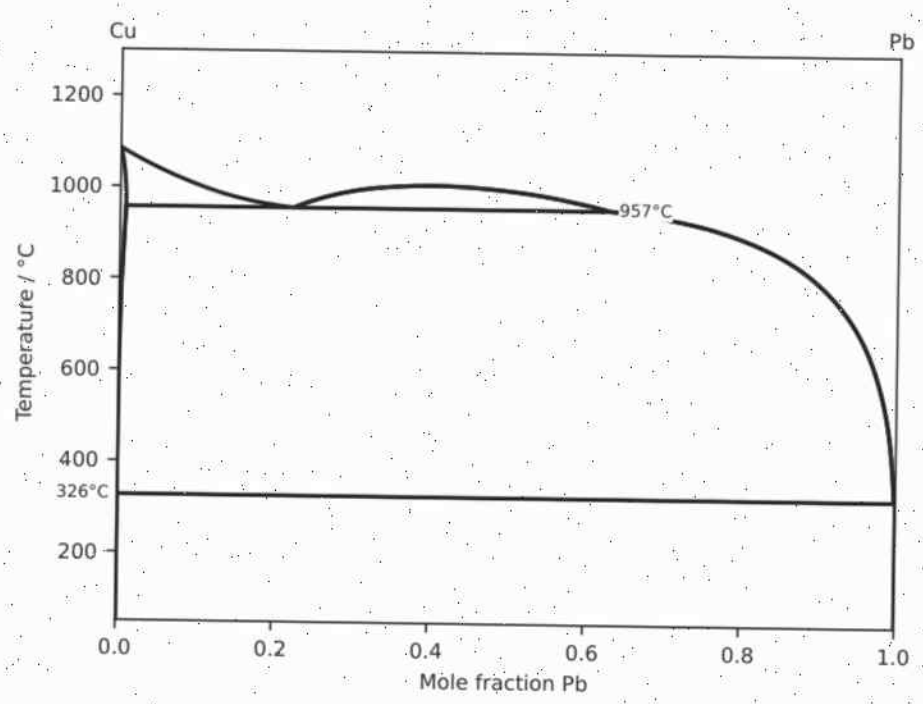


Fig. 1. Calculated Cu–Pb phase diagram.

At 956.8 °C the monotectic L1 → (Cu) + L2 takes place; the L1 phase then contains 22.2 at.% Pb. The eutectic reaction L → (Cu) + (Pb) is calculated at 326.3 °C, with L at 99.8 at.%.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Cu–Pb).

Reaction	Type	T / °C			Compositions (x(Pb))		
		this work	exp.	Ref.	phase 1	phase 2	phase 3
L1 → (Cu) + L2	monotectic	956.8	–	–	0.222	0.007	0.631
L → (Cu) + (Pb)	eutectic	326.3	321.3	[4]	0.998	0.000	1.000

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Cu–Pb).

Phase	Pearson	SB	Prototype	T min (K)	T max (K)	Range (x(Pb))	T (K)
(Cu)	cF4	A1	Cu	< 300	1357.8	0.000–0.490	600
HCP_A3	hP2	A3	Mg	599.4	600.6	0.150–0.750	600
(Pb)	cF4	A1	Cu	< 300	600.6	0.500–1.000	600

The upper stability limit of HCP_A3 is 327.5 °C. The solubility of Cu in (Pb) reaches 50.0 at.% at 326.9 °C.

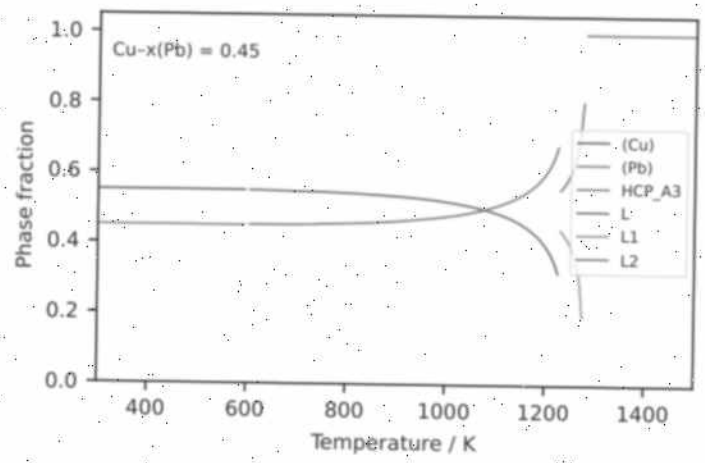


Fig. 2. Calculated equilibrium phase fractions versus temperature for Cu–x(Pb) = 0.45.

4. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its maximum, 6806 J/mol, at 48.0 at.% Pb. The calculated enthalpy of formation of HCP_A3 at 298.15 K is 0.42 kJ/mol per mole of atoms. Its heat capacity at room temperature is 25.86 J/(mol K). For the equiatomic liquid at 1450 K, the Gibbs energy of mixing is −3.09 kJ/mol-atom. In older units the corresponding mixing enthalpy is 1625 cal/mol (1 cal = 4.184 J).

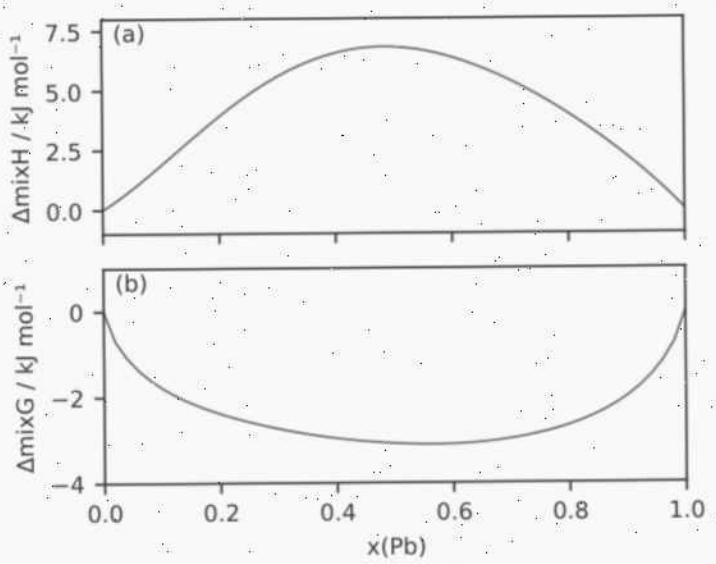


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1450 K.

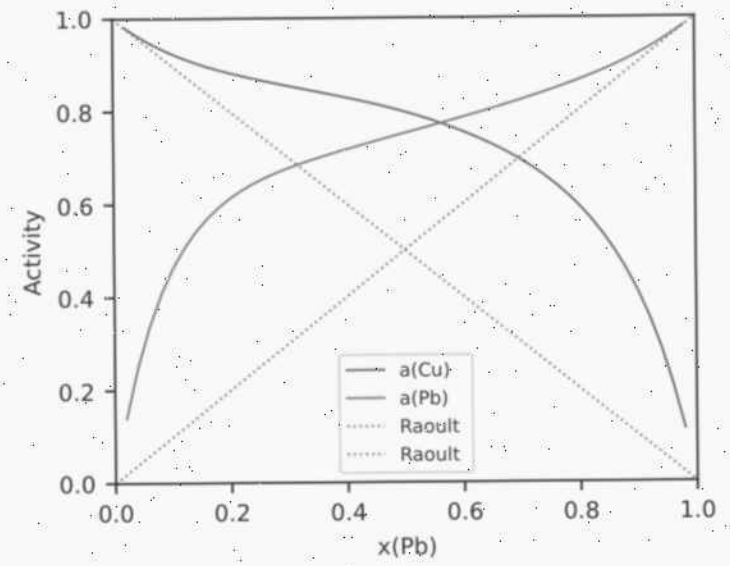


Fig. 4. Calculated activities in the liquid at 1450 K (reference: pure liquids).

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Cu–Pb).

Phase	at.% Pb	ΔfH(298.15 K) / cal mol⁻¹	ΔfS / cal mol⁻¹ K⁻¹	Cp(298.15 K) / cal mol⁻¹ K⁻¹
HCP_A3	59.0	101 ± 120	−0.16	6.18

Reference: stable elements (SER) at 298.15 K; values per mole of atoms; 1 cal = 4.184 J (thermochemical calorie). ± values are synthetic illustration uncertainties, not data.

Table 4. Integral and partial mixing properties of the liquid at 1450 K (Cu–Pb).

x(Pb)	ΔmixH / J mol⁻²	ΔmixG / J mol⁻¹	a(Cu)	a(Pb)	ΔH(Cu) / J mol⁻¹	ΔH(Pb) / J mol⁻¹
0.100	1862	−1782	0.9251	0.4599	−211	20518
0.200	3904	−2385	0.8827	0.6128	67	19250
0.300	5562	−2736	0.8545	0.6773	1508	15019
0.400	6551	−2962	0.8288	0.7170	4085	10250
0.500	6798	−3086	0.7978	0.7512	7355	6240
0.600	6365	−3101	0.7552	0.7854	10744	3446
0.700	5386	−2979	0.6931	0.8221	13832	1766
0.800	3987	−2661	0.5938	0.8645	16634	826
0.900	2223	−1988	0.4093	0.9195	19886	260

Reference states: pure liquid Cu and Pb at 1450 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] F.H. Hayes, H.L. Lukas, G. Effenberg, G. Petzow, Z. Metallkd., 77, 749-54(1986).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.